

Mathematics for Computer Science – Isomorphism, Homomorphism and Quotient Group.

Libin Wang

School of Computer Science, South China Normal University

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学习思路.

本章有一条联系紧密的知识链：从同态、同构出发，经过 Kernel、正规子群、商群等，最后到达同构定理。而这整条知识链围绕着一个核心目标：考察群结构。

Isomorphisms(同构.)

Motivation.

Many groups may have different appearances, however they are essentially same.

Isomorphisms(同构).

Definition of Isomorphism.

Two group $(\mathbb{G}, \cdot)$ and $(\mathbb{H}, \circ)$ are isomorphic if there exists a one-to-one and onto map $\phi : \mathbb{G} \mapsto \mathbb{H}$ such that the group operation is preserved; that is,

$$\phi(a \cdot b) = \phi(a) \circ \phi(b)$$

for all a and b in $\mathbb{G}$. If $\mathbb{G}$ is isomorphic to $\mathbb{H}$, we write $\mathbb{G} \cong \mathbb{H}$. The map ϕ is called an isomorphism.

Examples of Isomorphisms.

Example

$\mathbb{Z}_4 \cong \langle i \rangle$, since we can define a bijective map $\phi : \mathbb{Z}_4 \mapsto \langle i \rangle$ by $\phi(n) = i^n$. The map ϕ is one-to-one and onto, since

$$\phi(0) = 1$$

$$\phi(1) = i$$

$$\phi(2) = -1$$

$$\phi(3) = -i.$$

Moreover, ϕ preserves the group operation, since

$$\phi(m+n) = i^{m+n} = i^m i^n = \phi(m)\phi(n).$$

Examples of Isomorphisms.

Isomorphic groups.

Since $\mathbb{Z}_8^* = \{1, 3, 5, 7\}$, $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$, we can find an isomorphism ϕ to show that:

$$\mathbb{Z}_8^* \cong \mathbb{Z}_{12}^*$$

An isomorphism $\phi : \mathbb{Z}_8^* \mapsto \mathbb{Z}_{12}^*$ is defined by :

$$1 \mapsto 1$$

$$3 \mapsto 5$$

$$5 \mapsto 7$$

$$7 \mapsto 11$$

Can you find another isomorphism between these two groups?

(Question.)

Is $\mathbb{Z}_{61}^*$ isomorphic to $\mathbb{Z}_{77}^*$? Why or why not?

Examples of Isomorphisms.

Question-1.

Is $\mathbb{Z}_n$ isomorphic to $\mathbb{U}_n$? Why or why not?

Examples of Isomorphisms.

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Is $\mathbb{Z}_n$ isomorphic to $\mathbb{U}_n$? Why or why not?

Question-2.

Is $\mathbb{Z}_{61}^*$ isomorphic to $\mathbb{Z}_{77}^*$? Why or why not?

Theorem about Isomorphisms.

Proposition

Let $\phi : \mathbb{G} \mapsto \mathbb{H}$ be an isomorphism of two groups, then the following statements are true.

- 1 $\phi^{-1} : \mathbb{H} \mapsto \mathbb{G}$ is an isomorphism;
- 2 $|\mathbb{G}| = |\mathbb{H}|$;
- 3 If $\mathbb{G}$ is abelian, then $\mathbb{H}$ is abelian;
- 4 If $\mathbb{G}$ is cyclic, then $\mathbb{H}$ is cyclic;
- 5 if $\mathbb{G}$ has a subgroup of order n , then $\mathbb{H}$ has a subgroup of order n .

Proof.

Left as an exercise.



Theorem about Isomorphisms.

Theorem

All cyclic groups of infinite order are isomorphic to $\mathbb{Z}$.

Proof.

Suppose $\mathbb{G}$ is a cyclic group with infinite order, and $g \in \mathbb{G}$ is a generator. Define $\phi : \mathbb{Z} \mapsto \mathbb{G}$ by $\phi : n \mapsto g^n$. Then

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n).$$

Show ϕ is a bijective map. Left as an exercise.



Theorem about Isomorphisms.

Theorem

If $\mathbb{G}$ is a cyclic group of order n , then $\mathbb{G}$ is isomorphic to $\mathbb{Z}_n$.

Proof.

Let $\mathbb{G}$ be a cyclic group with order n , generated by g . Define $\phi : \mathbb{Z}_n \mapsto \mathbb{G}$ by $\phi : k \mapsto g^k$, where $0 \leq k < n$. Show ϕ is an isomorphism. Left as an exercise. □

Theorem about Isomorphisms.

Corollary

If $\mathbb{G}$ is a cyclic group of order p where p is a prime, then $\mathbb{G}$ is isomorphic to $\mathbb{Z}_p$.

Proof.

Easy!



单位根群.

$\mathbb{U}_n$ 与 $\mathbb{Z}_n$ 同构! 请问, 令 $n = p - 1$, $\mathbb{U}_n$ 与 $\mathbb{Z}_p^*$ 同构吗? 这个答案揭示了什么信息?

Theorem about Isomorphisms.

Theorem

The isomorphism of groups determines an equivalence relation on the class of all groups.

Proof.

Left as an exercise.



Theorem about Isomorphisms.

Theorem

(Cayley) Every group is isomorphic to a group of permutations.

Proof.

Omitted. Note that, it is important.



关于 Isomorphisms 的提醒.

提醒.

同构是研究群结构的目标!

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提问.

同态的目标是什么?

Homomorphisms. (同态)

Definition of Homomorphism.

Two group $(\mathbb{G}, \cdot)$ and $(\mathbb{H}, \circ)$ are homomorphic if there exists a map $\phi : \mathbb{G} \mapsto \mathbb{H}$ such that the group operation is preserved; that is,

$$\phi(a \cdot b) = \phi(a) \circ \phi(b)$$

for all a and b in $\mathbb{G}$. The map ϕ is called a homomorphism.

(Basic idea.)

We relax the requirement that an isomorphism of groups be bijective, we have a homomorphism.

Examples of Homomorphisms.

Example

$\mathbb{Z}$ 是加法群，定义映射 $\phi: \mathbb{Z} \mapsto \mathbb{Z}$ 为 $\phi(k) = 2k$, $\forall k \in \mathbb{Z}$ 。可以验证 ϕ 是一种群同态，因为

$$\phi(i+j) = 2(i+j) = 2i + 2j = \phi(i) + \phi(j)$$

ϕ 把整数映射为偶数。偶数在加法下成群。

Examples of Homomorphisms.

Example of Homomorphisms.

Let $\mathbb{G}$ be a group and $g \in \mathbb{G}$. Define a map $\phi : \mathbb{Z} \mapsto \mathbb{G}$ by $\phi(n) = g^n$. Then ϕ is a group homomorphism, since

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n).$$

This homomorphism maps $\mathbb{Z}$ onto the cyclic subgroup of $\mathbb{G}$ generated by g .

Examples of Homomorphisms.

Example

设 p 为素数, 定义映射 $\phi: \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$ 为 $\phi(g) = g^2, \forall g \in \mathbb{Z}_p^*$ 。可验证, ϕ 是一种同态映射, 因为对任意的 $g_1, g_2 \in \mathbb{Z}_p^*$ 满足:

$$\phi(g_1 g_2) = (g_1 g_2)^2 = g_1^2 g_2^2 = \phi(g_1) \phi(g_2)。$$

Normal subgroups (正规子群) .

Definition of normal subgroups.

A subgroup $\mathbb{H}$ of a group $\mathbb{G}$ is normal in $\mathbb{G}$ if $g\mathbb{H} = \mathbb{H}g$ for all $g \in \mathbb{G}$.

(Basic idea 1.)

A normal subgroup is a subgroup that the right cosets and the left cosets are precisely the same, and $g\mathbb{H} = \mathbb{H}g$ represents a kind of "commutativity(交换性)".

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A normal subgroup is a subgroup that the right cosets and the left cosets are precisely the same, and $g\mathbb{H} = \mathbb{H}g$ represents a kind of "commutativity(交换性)".

(Basic idea 2.)

A subgroup $\mathbb{H}$ of a group $\mathbb{G}$ is normal in $\mathbb{G}$ iff $\forall g \in \mathbb{G}$, $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$. Moreover, for all $\forall g \in \mathbb{G}$, $g\mathbb{H}g^{-1} = \mathbb{H}$

Basic Properties of Normal Subgroup.

Proposition

Let $\mathbb{G}$ be a group and $\mathbb{H}$ be a subgroup of $\mathbb{G}$. Then the following statements are equivalent.

- 1 The subgroup $\mathbb{H}$ is a normal subgroup of $\mathbb{G}$, namely, $g\mathbb{H} = \mathbb{H}g$ for all $g \in \mathbb{G}$.*
- 2 For all $g \in \mathbb{G}$, $g\mathbb{H}g^{-1} = \mathbb{H}$.*

Basic Properties of Normal Subgroup.

Proof.

(Proof of last proposition–1.) (1) $\Rightarrow$ (2). Since $\mathbb{H}$ is a normal subgroup of $\mathbb{G}$, $g\mathbb{H} = \mathbb{H}g$ for all $g \in \mathbb{G}$. Hence, for a given $g \in \mathbb{G}$ and $h \in \mathbb{H}$, there exists an $h' \in \mathbb{H}$ such that $gh = h'g$. Therefore, $ghg^{-1} = h' \in \mathbb{H}$ or $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$. For $h \in \mathbb{H}$, $g^{-1}hg = g^{-1}h(g^{-1})^{-1} \in \mathbb{H}$. Hence, $g^{-1}hg = h'$ for some $h' \in \mathbb{H}$. Therefore, $h = gh'g^{-1} \in g\mathbb{H}g^{-1}$, namely, $\mathbb{H} \subseteq g\mathbb{H}g^{-1}$.



Basic Properties of Normal Subgroup.

Proof.

(Proof of last proposition-2.)

(2) $\Rightarrow$ (1). Suppose that for all $g \in G$, $gHg^{-1} = H$. Then for any $h \in H$ there exists an $h' \in H$ such that $ghg^{-1} = h'$. Consequently, $gh = h'g$ which means $gH \subseteq Hg$. Similarly, we can prove that $Hg \subseteq gH$. □

Basic Properties of Homomorphisms.

Proposition

Proposition 1. Let $\phi : \mathbb{G}_1 \mapsto \mathbb{G}_2$ be a homomorphism of groups. Then

- ❶ *If e is the identity of $\mathbb{G}_1$, then $\phi(e)$ is the identity of $\mathbb{G}_2$;*
- ❷ *For any element $g \in \mathbb{G}_1$, $\phi(g^{-1}) = [\phi(g)]^{-1}$;*
- ❸ *If $\mathbb{H}_1$ is a subgroup of $\mathbb{G}_1$, then $\phi(\mathbb{H}_1)$ is a subgroup of $\mathbb{G}_2$;*
- ❹ *If $\mathbb{H}_2$ is a subgroup of $\mathbb{G}_2$, then $\phi^{-1}(\mathbb{H}_2)$ is a subgroup of $\mathbb{G}_1$. Furthermore, if $\mathbb{H}_2$ is normal in $\mathbb{G}_2$, then $\phi^{-1}(\mathbb{H}_2)$ is normal in $\mathbb{G}_1$.*

Basic Properties of Homomorphisms.

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- ❷ *For any element $g \in \mathbb{G}_1$, $\phi(g^{-1}) = [\phi(g)]^{-1}$;*
- ❸ *If $\mathbb{H}_1$ is a subgroup of $\mathbb{G}_1$, then $\phi(\mathbb{H}_1)$ is a subgroup of $\mathbb{G}_2$;*
- ❹ *If $\mathbb{H}_2$ is a subgroup of $\mathbb{G}_2$, then $\phi^{-1}(\mathbb{H}_2)$ is a subgroup of $\mathbb{G}_1$. Furthermore, if $\mathbb{H}_2$ is normal in $\mathbb{G}_2$, then $\phi^{-1}(\mathbb{H}_2)$ is normal in $\mathbb{G}_1$.*

Proof.

Omitted.



Basic Properties of Homomorphisms.

Definition

(Definition of Kernel.) Let $\phi : \mathbb{G} \mapsto \mathbb{H}$ be a group homomorphism and e is the identity of $\mathbb{H}$. By previous proposition, $\phi^{-1}(\{e\})$ is subgroup of $\mathbb{G}$. This subgroup is called the kernel of ϕ and denoted by $\text{Ker } \phi$.

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Example

设 p 为素数, 同态映射 $\phi : \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$ 定义为 $\phi(g) = g^2, \forall g \in \mathbb{Z}_p^*$.
可验证, ϕ 把 $\{1, p-1\}$ 映射为群 $\phi(\mathbb{Z}_p^*)$ 的单位元 1 , 所以
 $\text{Ker } \phi = \{1, p-1\}$.

Basic Properties of Homomorphisms.

Proposition

(Kernel.) Let $\phi : \mathbb{G} \mapsto \mathbb{H}$ be a group homomorphism. Then the kernel of ϕ is a normal subgroup of $\mathbb{G}$.

Basic Properties of Homomorphisms.

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(Kernel.) Let $\phi : \mathbb{G} \mapsto \mathbb{H}$ be a group homomorphism. Then the kernel of ϕ is a normal subgroup of $\mathbb{G}$.

Proof.

Trivial. Since the trivial subgroup of $\mathbb{H}$ is normal.



关于 Homomorphisms 的提醒.

提醒

同态是研究群结构的重要手段：通过群同态去找正规子群，即同态映射的 Kernel.

习题.

- 如果 $\mathbb{H}_1$ 和 $\mathbb{H}_2$ 是群 $\mathbb{G}$ 的正规子群, 证明 $\mathbb{H}_1\mathbb{H}_2$ 也是群 $\mathbb{G}$ 的正规子群。
- 证明, $\mathbb{H}$ 是 $\mathbb{G}$ 的正规子群当且仅当对任意 $g \in \mathbb{G}$, $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$ 。
- 定义映射 $\phi: \mathbb{G} \mapsto \mathbb{G}$ 为: $g \mapsto g^2$ 。请证明 ϕ 是一种群同态当且仅当 G 是阿贝尔群。
- 设 $\phi: \mathbb{G} \mapsto \mathbb{H}$ 是一种群同态。请证明: 如果 $\mathbb{G}$ 是循环群, 则 $\phi(\mathbb{G})$ 也是循环群; 如果 $\mathbb{G}$ 是交换群, 则 $\phi(\mathbb{G})$ 也是交换群。

Quotient Groups(商群).

Theorem

If $\mathbb{H}$ is a normal subgroup of a group $\mathbb{G}$, then the cosets of $\mathbb{H}$ in $\mathbb{G}$ form a group $\mathbb{G}/\mathbb{H}$ under the operation $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$. This group is call the quotient group or factor group of $\mathbb{G}$ and $\mathbb{H}$.

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Understand the operation.

How to understand the operation $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$?

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Understand the operation.

How to understand the operation $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$?

Since $\mathbb{H}$ is normal, then:

$$(a\mathbb{H})(b\mathbb{H}) = (\mathbb{H}a)(b\mathbb{H}) = (ab\mathbb{H})\mathbb{H} = ab\mathbb{H}$$

进一步理解商群的群操作.

Understand the operation.

To understand the operation $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$, we show that since $\mathbb{H}$ is normal, then:

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使用了两种属性:

- 正规子群的“交换性”
- 结合律

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使用了两种属性:

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另一种理解.

有没有另一种理解方式? 借助正常的思维 (什么是正常思维?) 与底层定义 (直接借助群元素的操作!).

进一步理解商群的群操作.

正常思维!

我们需要证明 $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$, 即我们需要证明:

- $(a\mathbb{H})(b\mathbb{H}) \subseteq ab\mathbb{H}$
- $ab\mathbb{H} \subseteq (a\mathbb{H})(b\mathbb{H})$

进一步理解商群的群操作.

正常思维!

我们需要证明 $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$, 即我们需要证明:

- $(a\mathbb{H})(b\mathbb{H}) \subseteq ab\mathbb{H}$
- $ab\mathbb{H} \subseteq (a\mathbb{H})(b\mathbb{H})$

Proof.

要证明 $(a\mathbb{H})(b\mathbb{H}) \subseteq ab\mathbb{H}$, 即任取 $ah_1bh_2 \in a\mathbb{H}b\mathbb{H}$, 证 $ah_1bh_2 \in ab\mathbb{H}$ 。但是, 这是容易的, 因为:

$$\begin{aligned} ah_1bh_2 &= h_3abh_2 \\ &= abh_4h_2 \\ &= abh_5 \in ab\mathbb{H} \end{aligned}$$

另一个方向, 证明 $ab\mathbb{H} \subseteq (a\mathbb{H})(b\mathbb{H})$ 则留给大家作为练习。 □

Theorem of Quotient Groups.

Theorem

(Quotient Groups). If $\mathbb{H}$ is a normal subgroup of a group $\mathbb{G}$, then the cosets of $\mathbb{H}$ in $\mathbb{G}$ form a group $\mathbb{G}/\mathbb{H}$ of order $[\mathbb{G} : \mathbb{H}]$.

Theorem of Quotient Groups.

Proof.

(Basic ideas.)

- 1 What is the group operation? $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$

Theorem of Quotient Groups.

Proof.

(Basic ideas.)

- 1 What is the group operation? $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$
- 2 Prove this operation is well-defined, i.e., the group operation must be independent of the choice of coset representative.
Let $a\mathbb{H} = b\mathbb{H}$, $c\mathbb{H} = d\mathbb{H}$. We must prove that

$$(a\mathbb{H})(c\mathbb{H}) = ac\mathbb{H} = bd\mathbb{H} = (b\mathbb{H})(d\mathbb{H})$$

Theorem of Quotient Groups.

Proof.

(Basic ideas.)

- 1 What is the group operation? $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$
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- 3 Why we need "well-defined"?

Theorem of Quotient Groups.

Proof.

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- 1 What is the group operation? $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$
- 2 Prove this operation is well-defined, i.e., the group operation must be independent of the choice of coset representative.
Let $a\mathbb{H} = b\mathbb{H}$, $c\mathbb{H} = d\mathbb{H}$. We must prove that

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- 3 Why we need "well-defined"?
- 4 Check the axioms of group. Easy!



Theorem of Quotient Groups.

Remark

(良定义操作.) 所谓良定义的操作, 就是要求操作独立于所参与操作的代表元。比如, 对任意群 $\mathbb{G}$ 和其上的某种操作 $\psi : \mathbb{G} \mapsto \mathbb{G}$, 要求 ψ 良定义就是要求对任意的群元 $a, b \in \mathbb{G}$, 如果 $a = b$, 则 $\psi(a) = \psi(b)$ 。一眼看上去, 这个要求很无理, 毫无意义, 但是对于商群来说就必不可少。请注意, 商群中操作的是陪集, $a\mathbb{H} = b\mathbb{H}$ 并不意味 $a = b$ 。

Theorem of Quotient Groups.

Remark

(群操作的良定义性.) Let $a\mathbb{H} = b\mathbb{H}$, $c\mathbb{H} = d\mathbb{H}$. We must prove that

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Theorem of Quotient Groups.

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$$(a\mathbb{H})(c\mathbb{H}) = ac\mathbb{H} = bd\mathbb{H} = (b\mathbb{H})(d\mathbb{H})$$

For $a = bn_1$ and $c = dn_2$ for some n_1 and n_2 in $\mathbb{H}$. Hence,

$$\begin{aligned}ac\mathbb{H} &= bn_1dn_2\mathbb{H} \\ &= bn_1d\mathbb{H} \\ &= bn_1\mathbb{H}d \\ &= b\mathbb{H}d \\ &= bd\mathbb{H}\end{aligned}$$

小结-商群的构造.

(构造商群的思路).

- 首先, 给定群 G , 找到它的一个正规子群 H , 比如通过同态映射的 Kernel 来找到一个正规子群。
- 其次, 构造商群的元素, 即通过 H 构造陪集。
- 第三, 定义商群的群操作, 即陪集的乘法, 并验证这个操作的良好定义性。
- 最后, 验证在所定义的乘法上, 商群的确实满足群公理: 结合律、单位元、逆元、封闭性。

Example for Quotient Groups.

(Example of Quotient Groups).

Consider the normal subgroup $3\mathbb{Z}$ of $\mathbb{Z}$. The cosets of $3\mathbb{Z}$ in $\mathbb{Z}$ are

$$0 + 3\mathbb{Z} = \{\dots, -3, 0, 3, 6, \dots\}$$

$$1 + 3\mathbb{Z} = \{\dots, -2, 1, 4, 7, \dots\}$$

$$2 + 3\mathbb{Z} = \{\dots, -1, 2, 5, 8, \dots\}.$$

The group $\mathbb{Z}/3\mathbb{Z}$ is given by the multiplicative table below.

+	$0 + 3\mathbb{Z}$	$1 + 3\mathbb{Z}$	$2 + 3\mathbb{Z}$
$0 + 3\mathbb{Z}$	$0 + 3\mathbb{Z}$	$1 + 3\mathbb{Z}$	$2 + 3\mathbb{Z}$
$1 + 3\mathbb{Z}$	$1 + 3\mathbb{Z}$	$2 + 3\mathbb{Z}$	$0 + 3\mathbb{Z}$
$2 + 3\mathbb{Z}$	$2 + 3\mathbb{Z}$	$0 + 3\mathbb{Z}$	$1 + 3\mathbb{Z}$

Example for Quotient Groups.

(Quotient Groups of $\mathbb{Z}_n^*$).

Let $n = 15$, then $\mathbb{Z}_n^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$. Let $g = 2$, we set $\mathbb{S} = \langle g \rangle = \{1, 2, 4, 8\}$ which is a subgroup of $\mathbb{Z}_n^*$. Then $\mathbb{Z}_n^*/7\mathbb{S} = \{\mathbb{S}, 7\mathbb{S}\}$, please check that $\mathbb{S}$ is the identity, $7\mathbb{S}$'s inverse is itself, namely $(7\mathbb{S})(7\mathbb{S}) = 4\mathbb{S} = \mathbb{S}$.

Theorem of Quotient Groups.

Example

(非良定义操作与商群) 设有限循环群 $G = \langle g \rangle$, H 是群 G 的子群, 自然也是正规子群, 于是形成商群 G/H 。定义映射 $\phi_g : G/H \mapsto \mathbb{Z}$, 对任意的 aH , 如果 $a = g^k$, 则 $\phi_g(aH) = k$ 。可以验证这是一种非良定义映射。

Theorem of Quotient Groups.

Example

(非良定义操作与商群) 比如, 设 $p = 7$, 则 $\mathbb{Z}_p^* = \{1, 2, 3, 4, 5, 6\}$, 取生成元 $g = 3$ 。记 $S = \{1, 6\}$, 可验证 S 是 $\mathbb{Z}_p^*$ 的子群。所以形成商群, $\mathbb{Z}_p^*/S = \{S, 2S, 3S\}$ 。此时 $3S = 4S$, $\phi_3(3S) = 1$, 但是 $\phi_3(4S) = 4$ 。也就是说, 对同一个陪集, 因为选取了不同的代表元, 得到的映射值不同, 该操作不能独立于代表元的选取。

思路.

商群怎么用？答：以上知识点可以形成这样一条知识链：从一个群同态 $\phi: \mathbb{G}_1 \mapsto \mathbb{G}_2$ 出发。首先可以找到 ϕ 的 Kernel，于是得到一个 $\mathbb{G}_1$ 的正规子群；接着，构造 $\mathbb{G}_1$ 上的商群 $\mathbb{G}_1 / \text{Ker } \phi$ 。再通过“同构定理”完成一直以来的主要任务：研究群（商群）的结构！)

Canonical Homomorphism.

正规同态（自然同态）.

Let $\mathbb{H}$ be a normal subgroup of $\mathbb{G}$, define a map

$$\phi : \mathbb{G} \mapsto \mathbb{G}/\mathbb{H}$$

by

$$\phi(g) = g\mathbb{H}.$$

This map is indeed a homomorphism, check it! We call this map a natural or canonical homomorphism, and $\text{Ker } \phi = \mathbb{H}$.

First Isomorphism Theorem.

Theorem

(First Isomorphism Theorem.) If $\psi : \mathbb{G} \mapsto \mathbb{H}$ is a group homomorphism with $\mathbb{K} = \text{Ker } \psi$, then $\mathbb{K}$ is normal in $\mathbb{G}$. Let $\phi : \mathbb{G} \mapsto \mathbb{G}/\mathbb{K}$ be the canonical homomorphism. Then there exists a unique isomorphism $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$ such that $\psi = \eta\phi$.

First Isomorphism Theorem.

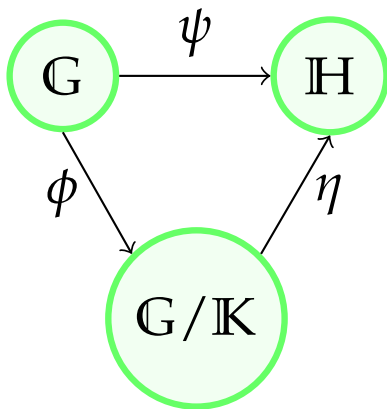


Figure: A diagrammatic interpretation of First Isomorphism Theorem.

关于第一同构定理的直观认识.

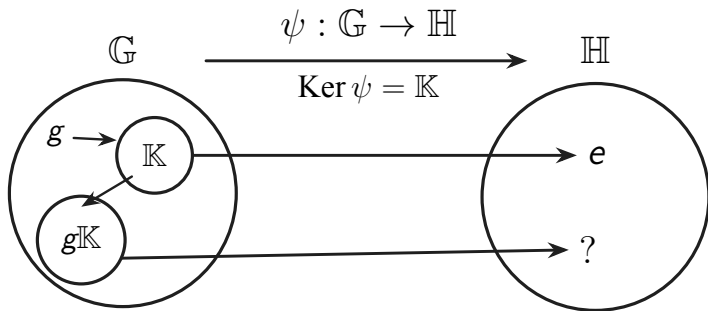


Figure: 关于第一同构定理的直观认识.

First Isomorphism Theorem.

(证明思路.)

- 1 Define $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$ by $\eta(g\mathbb{K}) = \psi(g)$;
- 2 Prove η is well-defined;
- 3 Prove that η is a homomorphism and is a bijective map.

First Isomorphism Theorem.

(证明第一步.)

- 定义映射 $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$ 为 $\eta(g\mathbb{K}) = \psi(g)$ 。

First Isomorphism Theorem.

(证明第二步.)

- 证明 η 是良定义映射, 即证明如果对于两个不同的 $g_1, g_2 \in \mathbb{G}$, 有 $g_1\mathbb{K} = g_2\mathbb{K}$, 则 $\eta(g_1\mathbb{K}) = \eta(g_2\mathbb{K})$ 。因为 $g_1\mathbb{K} = g_2\mathbb{K}$, 则存在 $k \in \mathbb{K}$ 使得 $g_1 = g_2k$, 这仅仅利用了陪集相等的定义。因此:

$$\eta(g_1\mathbb{K}) = \psi(g_1) = \psi(g_2k) = \psi(g_2)\psi(k) = \psi(g_2) = \eta(g_2\mathbb{K})$$

请注意, 上式中第四个等号之所以成立是因为 $k \in \mathbb{K}$, 而 $\mathbb{K}$ 是 ψ 的 Kernel, k 必然映射到 $\mathbb{H}$ 的单位元。这就证明了 η 映射独立于陪集代表元的选择。之所以 η 是唯一定义的, 因为给定了 ψ 和 ϕ , 而 $\psi = \eta\phi$, 且 ϕ 是满射。

First Isomorphism Theorem.

(一个问题.)

确保 η 的唯一性, 为什么需要 ϕ 是满射? (课后练习!)

First Isomorphism Theorem.

(证明第三步.)

- 可证明 η 是同态且是双射。 η 是同态, 因为:

$$\begin{aligned}\eta(g_1\mathbb{K}g_2\mathbb{K}) &= \eta(g_1g_2\mathbb{K}) \\ &= \psi(g_1g_2) \\ &= \psi(g_1)\psi(g_2) \\ &= \eta(g_1\mathbb{K})\eta(g_2\mathbb{K})\end{aligned}$$

η 显然是满射, 最后, 只需证明 η 是单射。

First Isomorphism Theorem.

(证明第三步.)

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η 显然是满射， 最后， 只需证明 η 是单射。

(一点点提醒.)

为什么 η 显然是满射？

First Isomorphism Theorem.

(证明第三步.)

- 证明 η 是单射。任取 $g_1\mathbb{K}, g_2\mathbb{K}$, 假设 $\eta(g_1\mathbb{K}) = \eta(g_2\mathbb{K})$, 则 $\psi(g_1) = \psi(g_2)$ 。 g_1 和 g_2 是 $\mathbb{G}$ 中两个不同的元素, 所以

$$\psi(g_1) = \psi(g_2 g_2^{-1} g_1) = \psi(g_2) \psi(g_2^{-1} g_1)$$

即 $\psi(g_2^{-1} g_1)$ 是 $\psi(\mathbb{G})$ 的单位元, 也即 $g_2^{-1} g_1 \in \mathbb{K}$ 。所以 $g_2^{-1} g_1 \mathbb{K} = \mathbb{K}$, 最后得到 $g_1 \mathbb{K} = g_2 \mathbb{K}$ 。

Example for First Isomorphism Theorem.

(Homomorphism from Cyclic Group.)

设 $\mathbb{G}$ 是由生成元 g 生成的循环群。定义映射 $\phi: \mathbb{Z} \mapsto \mathbb{G}$ 为 $n \mapsto g^n, \forall n \in \mathbb{Z}$ 。 ϕ 是同态映射，因为：

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n)。$$

ϕ 显然是满射。如果 $\mathbb{G}$ 的阶为 m ，因为 g 是生成元，则 $\text{ord}(g) = m$ 。于是， $g^m = e$ ，且有 $\text{Ker } \phi = m\mathbb{Z}$ 。根据第一同构定理，则有：

$$\mathbb{Z} / \text{Ker } \phi = \mathbb{Z} / m\mathbb{Z} \cong \mathbb{G}。$$

如果 $\mathbb{G}$ 是无限阶，则 g 也是无限阶，则 $\text{Ker } \phi = \{0\}$ ，则 $\mathbb{Z}$ 与 $\mathbb{G}$ 同构。因此，两个循环群同构当且仅当它们有相同的阶。在同构的意义上，只有两种循环群： $\mathbb{Z}$ 和 $\mathbb{Z}_n$ 。

Example for First Isomorphism Theorem.

(Homomorphism from $\mathbb{Z}_p^*$ to $\mathbb{Z}_p^*$.)

Let p be a prime, $\mathbb{Z}_p^*$ is a cyclic group. Define a map $\phi : \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$ by $\phi(g) = g^2$ for all $g \in \mathbb{Z}_p^*$. Then ϕ is a group homomorphism, since

$$\phi(g_1 g_2) = (g_1 g_2)^2 = g_1^2 g_2^2 = \phi(g_1) \phi(g_2).$$

Clearly ϕ is not onto, and $\text{Ker } \phi = \{1, p-1\}$ is a normal subgroup of $\mathbb{Z}_p^*$. We know $\text{Ker } \phi$ because we believe that the following equation

$$x^2 \equiv 1 \pmod{p}$$

has only two solutions, namely 1 and $p-1$. Check that $\mathbb{S} = \{\phi(g) : \text{for all } g \in \mathbb{Z}_p^*\}$ is a group. What is the order of $\mathbb{S}$? By the First Isomorphism Theorem, $|\mathbb{S}| = |\mathbb{Z}_p^* / \text{Ker } \phi| = |\mathbb{Z}_p^*| / |\text{Ker } \phi|$.

Example for First Isomorphism Theorem.

(Homomorphism from $\mathbb{Z}_n^*$ to $\mathbb{Z}_n^*$.)

Let $n = pq$ be a composite integer, p and q are two primes, and $\mathbb{Z}_n^*$ is a group. Define a map $\phi : \mathbb{Z}_n^* \mapsto \mathbb{Z}_n^*$ by $\phi(g) = g^2$ for all $g \in \mathbb{Z}_n^*$. Then ϕ is a group homomorphism. $S = \{\phi(g) : \text{for all } g \in \mathbb{Z}_n^*\}$, if we know the order of $\text{Ker } \phi$, then we know the order of $S = |\mathbb{Z}_n^*| / |\text{Ker } \phi|$ by the First Isomorphism Theorem. How many solutions does the following equation have?

$$x^2 \equiv 1 \pmod{n}$$

Unfortunately, we don't solve it until we learn CRT.

一个值得记忆的结论.

一个好用的结论.

设 $\phi: \mathbb{G} \mapsto \mathbb{H}$ 是一种群同态。 ϕ 是单射当且仅当 $\text{Ker } \phi = \{e\}$

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一个好用的结论.

设 $\phi: \mathbb{G} \mapsto \mathbb{H}$ 是一种群同态。 ϕ 是单射当且仅当 $\text{Ker } \phi = \{e\}$

证明.

- $\Rightarrow$: ϕ 是单射, 显然 $\text{Ker } \phi = \{e\}$ 。
- $\Leftarrow$: $\text{Ker } \phi = \{e\}$, 根据第一同构定理, 有 $\mathbb{G} \cong \mathbb{G} / \text{Ker } \phi \cong \phi(\mathbb{G})$ 。所以 ϕ 是单射。

例子-1.

同构.

设 G 和 H 是群。证明 $G \times H$ 同构于 $H \times G$

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证明.

- 思路：构造第一同构定理的“三角关系”。并使得 $\text{Ker } \phi = \{e\}$ 。

例子-2.

同构.

设 H 和 K 是群 G 的正规子群, 且 $H \cap K = \{e\}$. 证明 G 同构于 $G/H \times G/K$ 的一个子群。

例子-2.

同构.

设 H 和 K 是群 G 的正规子群, 且 $H \cap K = \{e\}$ 。证明 G 同构于 $G/H \times G/K$ 的一个子群。

证明.

- 思路: 构造第一同构定理的“三角关系”。并使得 $\text{Ker } \phi = \{e\}$ 。

Second Isomorphism Theorem.

Theorem

(第二同构定理.) H 是群 G 的子群 (不必然是正规子群), K 是群 G 的正规子群。则 HK 是群 G 的子群, $H \cap K$ 是 H 的正规子群, 且

$$H/(H \cap K) \cong HK/K.$$

Correspondence Theorem.

Correspondence Theorem. (对应定理)

Let $\mathbb{H}$ be a normal subgroup of a group $\mathbb{G}$. Then $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$ is a one-to-one correspondence between the set of subgroups $\mathbb{H}$ containing $\mathbb{H}$ and the set of subgroups of $\mathbb{G}/\mathbb{H}$. Furthermore, the normal subgroups of $\mathbb{G}$ containing $\mathbb{H}$ correspond to normal subgroups of $\mathbb{G}/\mathbb{H}$.

Correspondence Theorem.(对应定理)

Understanding Correspondence Theorem.

- 1 What is the map $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$?

Correspondence Theorem.(对应定理)

Understanding Correspondence Theorem.

- 1 What is the map $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$?
- 2 A map: $\{\text{the set of subgroups } \mathbb{H} \text{ containing } \mathbb{H}\} \mapsto \{\text{the set of subgroups of } \mathbb{G}/\mathbb{H}\}$

Correspondence Theorem.(对应定理)

Understanding Correspondence Theorem.

- ① What is the map $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$?
- ② A map: $\{\text{the set of subgroups } \mathbb{H} \text{ containing } \mathbb{H}\} \mapsto \{\text{the set of subgroups of } \mathbb{G}/\mathbb{H}\}$
- ③ To understand what is a subgroup of $\mathbb{G}/\mathbb{H}$?

Correspondence Theorem.

Proof ideas of the Correspondence Theorem.

- 1 $\mathbb{H}/\mathbb{H}$ is a subgroup of $\mathbb{G}/\mathbb{H}$;
- 2 The map $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$ is one-to-one and onto;
- 3 $\mathbb{H}$ is normal in $\mathbb{G}$, if and only if $\mathbb{H}/\mathbb{H}$ is normal in $\mathbb{G}/\mathbb{H}$.

Third Isomorphism Theorem.

Theorem

(第三同构定理) H 和 K 是群 G 的正规子群, 且 $K \subseteq H$ 。则:

$$G/H \cong \frac{G/K}{H/K}.$$

习题.

- 证明：如果 H 是群 G 上指标为 2 的子群，则 H 是 G 的正规子群。
- 给定任意群 G ， H 是群 G 的正规子群。请证明，如果群 G 是阿贝尔群，则商群 G/H 也是阿贝尔群。
- 给定任意群 G ， H 是群 G 的正规子群。请证明，如果群 G 是循环群，则商群 G/H 也是循环群。
- 设 p 和 q 是两个不同的素数， G 是阶为 pq 的阿贝尔群，请证明 G 是循环群。

习题.

设 p 和 q 是两个不同的素数, $\mathbb{G}$ 是阶为 pq 的阿贝尔群, 请证明 $\mathbb{G}$ 是循环群。

思路: 找阶为 p 的群元 g_1 和阶为 q 的群元 g_2 , 则可知 g_1g_2 的阶是 pq , 则 $\mathbb{G}$ 是循环群。

习题讲解.

习题.

设 p 和 q 是两个不同的素数, $\mathbb{G}$ 是阶为 pq 的阿贝尔群, 请证明 $\mathbb{G}$ 是循环群。

思路: 找阶为 p 的群元 g_1 和阶为 q 的群元 g_2 , 则可知 g_1g_2 的阶是 pq , 则 $\mathbb{G}$ 是循环群。

Proof.

不妨假设存在阶为 p 的群元 g_1 , $\mathbb{H} = \langle g_1 \rangle$ 为循环群。可得阶为 q 的商群 $\mathbb{G}/\mathbb{H}$, 它也必然为循环群。记 $\mathbb{G}/\mathbb{H}$ 的生成元是 $g_2\mathbb{H}$ 。此时, g_2 的阶不能是 1。同时, g_2 的阶不能是 p , 否则 $(g_2\mathbb{H})^p = \mathbb{H}$ 。如果 $p < q$, 则 $g_2\mathbb{H}$ 不能是生成元。如果 $p > q$, 则 q 整除 p , 这是不可能的! 所以, g_2 的阶只能是 q 或者 pq , 无论是什么情况, $\mathbb{G}$ 都是循环群。□

关于第一同构定理运用的习题.

习题.

- 证明: $\mathrm{GL}_n(R)/\mathrm{SL}_n(R)$ 与乘法群 $\mathbb{R}^*$ 同构。
- 证明商群 $\mathbb{R}/\mathbb{Z}$ 同构于圆圈群 $\mathbb{O}$ 。
- 设 $\phi: \mathbb{G} \mapsto \mathbb{H}$ 是一种群同态。请证明 ϕ 是单射当且仅当 $\mathrm{Ker} \phi = \{e\}$ 。
- 给定有限阿贝尔群 $\mathbb{G}$ 和正整数 n , 定义映射 $\phi: \mathbb{G} \mapsto \mathbb{G}$ 为, $\forall g \in \mathbb{G}, \phi(g) = g^n$ 。请分析并证明, 在什么条件下 ϕ 会是一种单射?